

Putnam 1985 — B1

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Let k be the smallest positive integer for which there exist distinct integers m_1, m_2, m_3, m_4, m_5 such that the polynomial

$$p(x) = (x - m_1)(x - m_2)(x - m_3)(x - m_4)(x - m_5)$$

has exactly k nonzero coefficients. Find, with proof, a set of integers m_1, m_2, m_3, m_4, m_5 for which this minimum k is achieved.

Solution

The answer is $k = 3$. It is achievable with $(m_1, m_2, m_3, m_4, m_5) = (0, -1, 1, -2, 2)$, where $p(x) = x^5 - 5x^3 + 4x$.

Firstly we show that $k \neq 1$. Observe that $\deg p(x) = 5$ and $p(x)$ is monic, so $p(x)$ must include the term x^5 . FTSOC assume $k = 1$. Then $p(x) \equiv x^5$. But this is a contradiction since we assumed that p had 5 distinct roots.

Next we show that $k \neq 2$. One of the non-zero coefficients has to be x^5 .

- If $p(x) = x^5 - c$, then it can be checked that $c^{\frac{1}{5}} e^{\frac{2\pi i}{5}}$ is a root of p . This is a contradiction since we assumed all the roots of p were integers.
- If $p(x) = x^5 - cx$, then it can be checked that $c^{\frac{1}{4}} i$ is a root of p . This again is a contradiction.
- If $c \neq 0$ is the coefficient of x^m , for $2 \leq m \leq 4$ an integer, then p will have a factor of x^2 , that is a repeated root of 0, contradicting the assumption that p had five distinct roots. ■